CMSC 27100 Practice Problems
Week #7

# Counting

Combinations, Permutations, Combinations with Replacement

1. How many distinct ways to form 2 pairs?
2. How many ways to rearrange the letters of the word COFFEE?
3. Given a collection of 12 students. How many different teams of 3 can we make?
4. At a supermarket, Sally wants to buy 12 pieces of fruit. She is able to pick from oranges, pears, and apples. How many different ways is there is buy 12 pieces of fruit?

# Binomial Theorem

Vandermonde, Pascal's Identity.

1. Calculate using Pascal's Identity:

$$\binom{11}{4}$$

$$\binom{12}{5}$$

$$\binom{15}{2}$$

2. Calculate using Vandermonde's Identity

$$\sum_{k=0}^{6}\binom{8}{k}\binom{5}{6-k}$$

$$\sum_{k=0}^{4}\binom{9}{k}\binom{7}{4-k}$$

$$\sum_{k\in\mathbb{Z}}\binom{12}{k}\binom{9}{10-k}$$